Spectrum Awareness for Cyber Operators
Why EMS matters in Cyber Defense

Problem

Modern military cyber operations depend on wireless communications, navigation, sensing, and unmanned systems operating within the electromagnetic spectrum (EMS). While doctrine recognizes EMS as a contested warfighting domain, most cyber defense training and tools remain focused on OSI Layers 3–7. Adversaries exploit this gap by targeting OSI Layers 1–2 through jamming, spoofing, and spectrum denial - disrupting communications before traditional cyber defenses can respond.

The Result

•Degraded command and control (C2)
•Loss of situational awareness
•Compromised UAS and sensor reliability
•Reduced mission effectiveness in contested environments

Core Gap

Cyber operators lack operational spectrum awareness integrated into their defensive workflows.

Key Operational Insights

•EMS superiority is a prerequisite for cyber resilience
•Spectrum control enables adversaries to bypass network perimeters
•Physical-layer attacks precede & neutralize higher-layer defenses
•Integrated cyber + EMS defense restores operational continuity
•Cyber defense w/o spectrum awareness is structurally incomplete

Technical Approach: Cross-Layer Defensive Architecture

We implemented a cross-layer defensive framework that secures mission-critical RF systems by combining spectrum monitoring with hardware safeguards and protocol-level countermeasures to detect, attribute, and mitigate physical-layer threats.

Core Components

• EMS Monitoring – Continuous SDR-based scanning for real-time threat detection, signal classification, and geolocation.
• RF Fingerprinting – Physical-layer authentication using hardware signatures to mitigate spoofing and unauthorized transmitters.
• Hardened MAC Protocols – Layer 2 defensive logic to mitigate protocol exploitation and denial mechanisms.

This architecture enables detection & mitigation before mission-layer degradation occurs

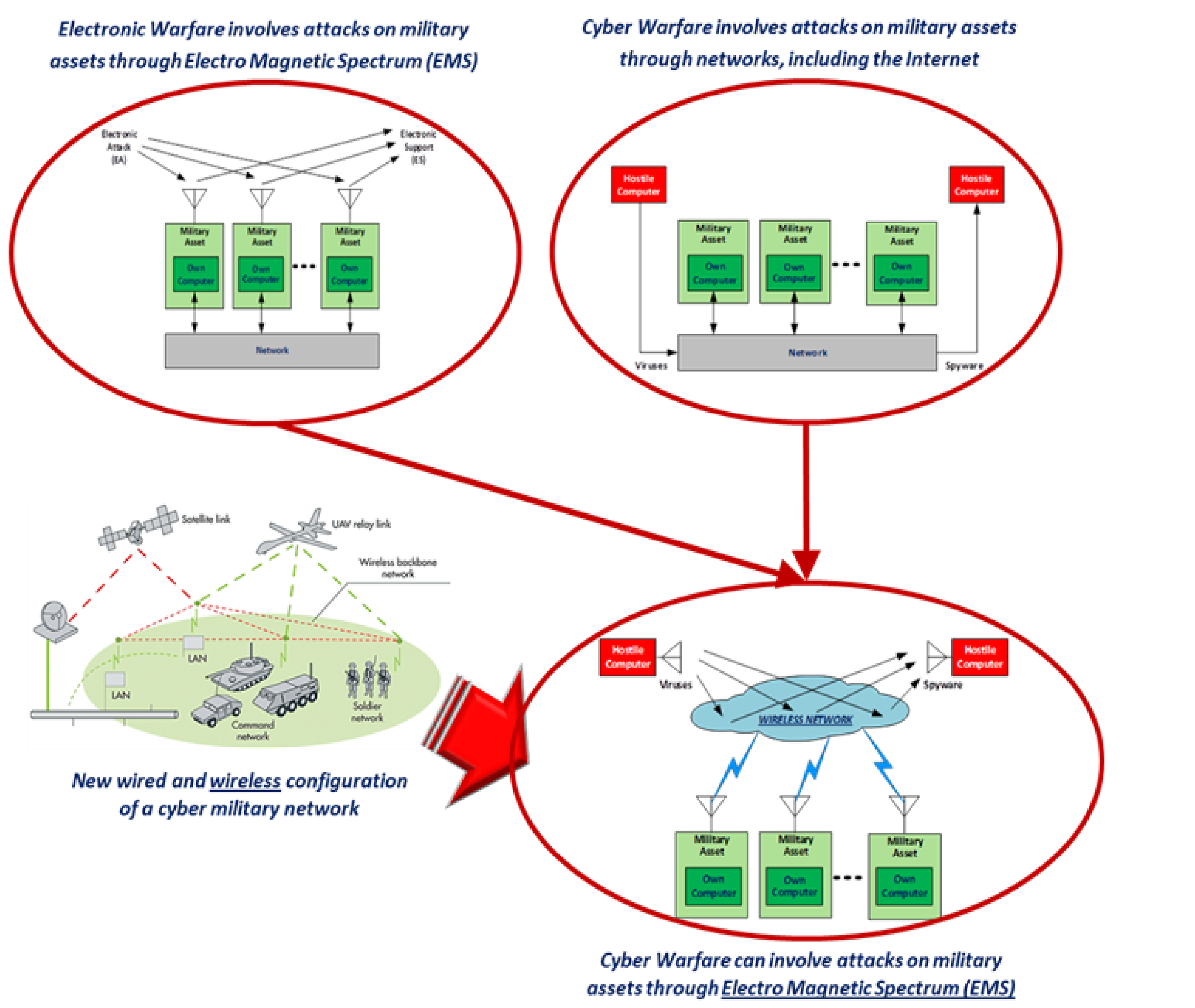

Figure 1: Modern cyber operations span both network and electromagnetic spectrum attack surfaces. Source: https://www.emsopedia.org/entries/communication-electronic-attack/

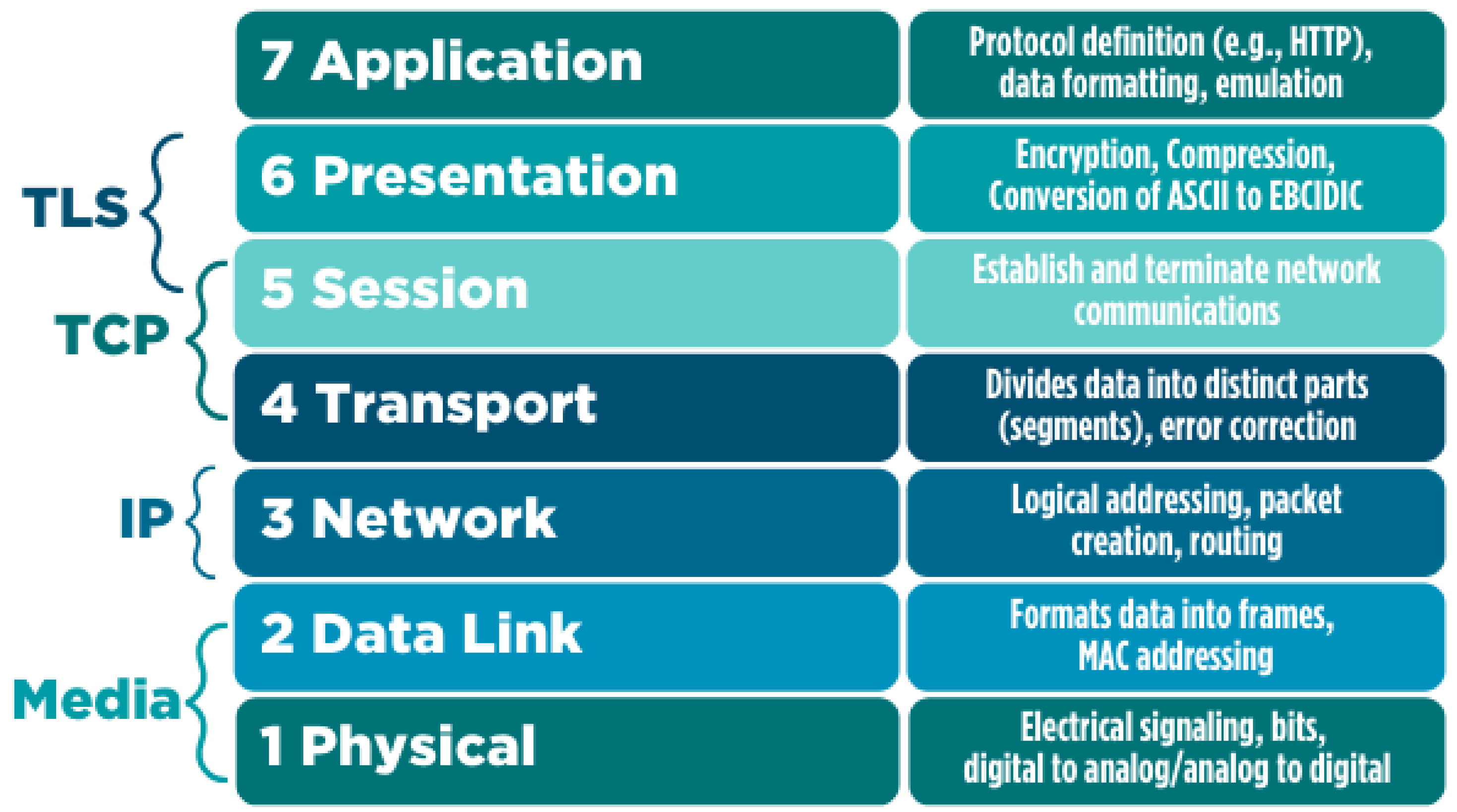

Figure 2. OsI model highlighting EMS attack surfaces at layers 1-2. Source: https://www.esecurityinstitute.com/the-osi-model/

The HCC Team

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Results & Operational Validation

•Confirmed the electromagnetic spectrum as a primary attack surface enabling adversaries to bypass traditional network defenses.
•Demonstrated that integrating EMS awareness into cyber workflows improves resilience of RF-dependent systems in contested environments.
•Developed a cross-layer framework linking EMS disruption directly to mission degradation across UAS, sensors, radios, and C2 systems.
•Established a foundation for spectrum-informed cyber defense training modules.

Next Phase – VICEROY Alignment

•Develop modular training content integrating EMS awareness into cyber operator curricula.
•Build contested-spectrum simulation environments to test link-layer mitigations (frequency hopping, adaptive MAC, physical-layer authentication).
•Prototype a spectrum-awareness decision workflow for cyber operators:
Detect → Attribute → Mitigate → Restore Mission Capability.

Establish a replicable training model scalable across VICEROY institutions.

Relevance to VICEROY Symposium 2026

This work advances:
•Cyber-EMS integration research
•Workforce development in contested-spectrum operations
•Applied training frameworks bridging academic research and operational readiness
•Scalable community-college-to-research-university talent pipelines

Houston City Collee proposes to operationalize EMS-aware cyber defense training at scale within the Department's defense ecosystem, n coordination w/VICEROY institutions

References

- EMSopedia – *Cyber Electromagnetic Activities*
- AFDP 3-85 – *Electromagnetic Spectrum Operations*
- DoD EMS Superiority Strategy (2020) – *Freedom of Action in the electromagnetic spectrum.*

Military Education Programs

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